ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Mechatronics paper 1 · practice d.

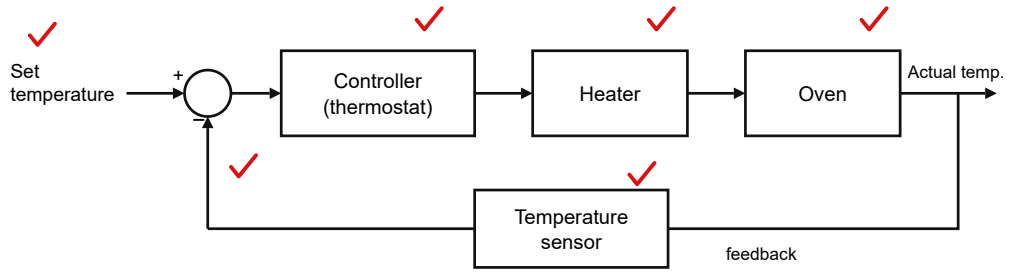
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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and wording. Symbols, circuits and diagrams may be drawn in different but correct ways.

QUESTION 1

1.1 A design brief is a short, clear statement of the problem and of what the product must do ✓✓. Any THREE ✓✓✓: the problem / need; who will use it; specifications (size, cost, materials); time allowed; constraints; how it will be tested. (5)

1.2



Set (desired) temperature as input ✓; controller (thermostat / control unit) ✓; heater ✓; oven (output temperature) ✓; temperature sensor in the feedback path ✓; arrows in the correct direction and comparison point ✓ (6)

1.3 Part of the output is measured and sent back to the controller ✓, which compares it with the desired value and corrects any difference ✓. (2)

1.4 Any TWO ✓✓: washing machine; microwave oven; robot vacuum cleaner; DVD player / printer; automatic gate motor; air conditioner. (2)

1.5 A manual system needs a person to operate and control it ✓, e.g. a hand drill / a light switch ✓. An automatic system controls itself using sensors and a controller without a person ✓✓, e.g. an automatic door / a thermostat-controlled geyser ✓. (5)

[20]

QUESTION 2

2.1.1 $\text{MA} = \text{load} \div \text{effort} = 800 \div 250 \checkmark = 3{,}2 \checkmark$ (2)

2.1.2 $\eta = \text{MA} \div \text{VR} \times 100 \checkmark = 3{,}2 \div 4 \times 100 \checkmark = 80\ \% \checkmark$ (3)

2.2.1 $\text{Gear ratio} = \text{T}_{\text{driven}} \div \text{T}_{\text{driver}} = 80 \div 20 \checkmark = 4\ (4:1) \checkmark$ (2)

2.2.2 $\text{N}_{\text{out}} = 1\,500 \div 4 \checkmark = 375\ \text{r/min} \checkmark$ (2)

2.2.3 $\text{T}_{\text{out}} = 2 \times 4 \checkmark = 8\ \text{N·m} \checkmark$ (2)

2.3.1 $\text{W} = \text{F} \times \text{s} = 50 \times 20 \checkmark = 1\,000\ \text{J} \checkmark$ (2)

2.3.2 $\text{P} = \text{W} \div \text{t} = 1\,000 \div 10 \checkmark = 100\ \text{W} \checkmark$ (2)

2.4 Any THREE ✓✓✓: spur gears; helical gears; bevel gears; worm and worm wheel; rack and pinion; internal gears. (3)

2.5 It stores rotational (kinetic) energy ✓ and keeps the speed of a shaft steady (smooths out speed changes) ✓. (2)

[20]

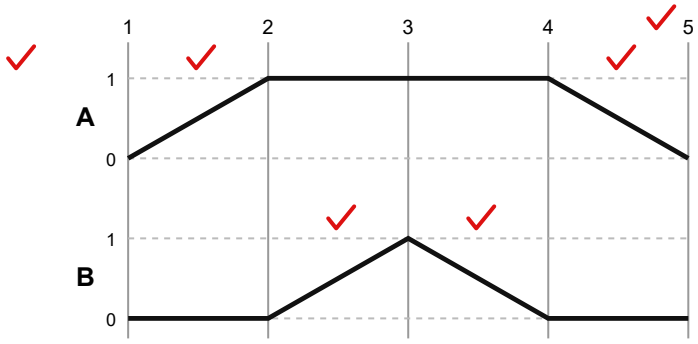
QUESTION 3

3.1.1 AND function ✓ (1)

3.1.2 The press only works when the operator presses BOTH valves (one with each hand) ✓, so the operator's hands cannot be under the press when it comes down ✓. (2)

3.1.3 Two-pressure valve (AND valve) ✓ (1)

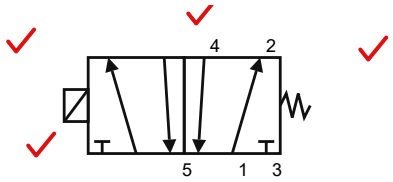
3.2



Two lines labelled A and B with 0 and 1 levels ✓; steps 1–5 marked ✓; A rises in step 1–2 ✓; B rises in step 2–3 ✓; B falls in step 3–4 ✓; A falls in step 4–5 ✓

(6)

3.3



Two squares with 5 ports ✓; flow paths correct in both positions ✓; solenoid actuator ✓; spring return ✓

(4)

3.4 The valve is placed where the cylinder rod (or a cam on it) presses its roller at the end of the stroke ✓; the valve then switches ✓ and sends a signal to the next step of the circuit (e.g. to retract the cylinder or start the next cylinder) ✓. (3)

3.5 To reduce the noise of the air leaving the valve ✓✓, protecting the hearing of workers (and keeping dirt out of the exhaust port) ✓. (3)

[20]

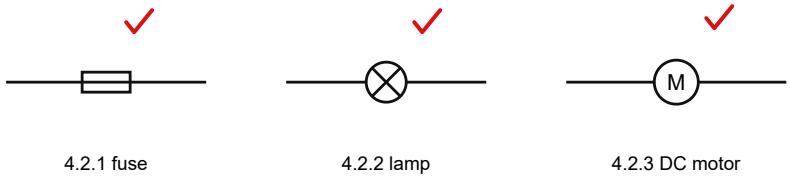
QUESTION 4

4.1.1 S0 – emergency stop ✓; S1 – STOP button ✓; S2 – START button ✓; F1 – overload relay contact ✓; K1 – contactor (relay) coil ✓ (5)

4.1.2 The F1 contact opens ✓, so coil K1 de-energises and the motor stops (the holding contact opens, so it does not restart on its own) ✓. (2)

4.1.3 That the motor is running (K1 is energised) ✓ (1)

4.2



4.2.1 fuse 4.2.2 lamp 4.2.3 DC motor

4.2.1 Fuse ✓✓ 4.2.2 Circle with a cross ✓✓ 4.2.3 Circle with M (and the supply leads) ✓✓ (6)

4.3 A short circuit is a very low-resistance path between live and neutral (or + and −) ✓ so that a very large current flows ✓. Protection: a fuse ✓ or circuit breaker ✓ that opens the circuit quickly. (4)

4.4 Any TWO ✓✓: if a live wire touches the frame, the current flows to earth and trips the protection; prevents electric shock to people touching the frame; prevents fires. (2)

[20]

QUESTION 5

5.1 Ferrous, any TWO ✓✓: mild steel; cast iron; stainless steel; tool steel. Non-ferrous, any TWO ✓✓: copper; aluminium; brass; lead; tin. (4)

5.2.1 Copper: good conductor of electricity / ductile ✓ – wires, PCB tracks ✓

5.2.2 Aluminium: light and corrosion-resistant ✓ – frames, heat sinks, profiles ✓

5.2.3 Nylon: tough, low friction, insulator ✓ – gears, bushes, cable ties ✓

(6)

5.3 Any FOUR ✓✓✓✓: use the right tool for the job; keep tools in good condition (no loose handles, sharp blades); cut away from your body; carry sharp tools pointing down; use insulated tools on electrical work; store tools properly after use; wear the correct PPE. (4)

5.4 Condition monitoring is checking the condition of a machine regularly while it is running ✓✓ to find signs of a problem early, so that repairs can be planned before it breaks down ✓✓. Any TWO examples ✓✓: vibration measurement; temperature checks (infrared thermometer); oil analysis; listening for noise; checking motor current; checking air pressure and leaks. (6)

[20]

TOTAL: 100

END OF MARKING GUIDELINE